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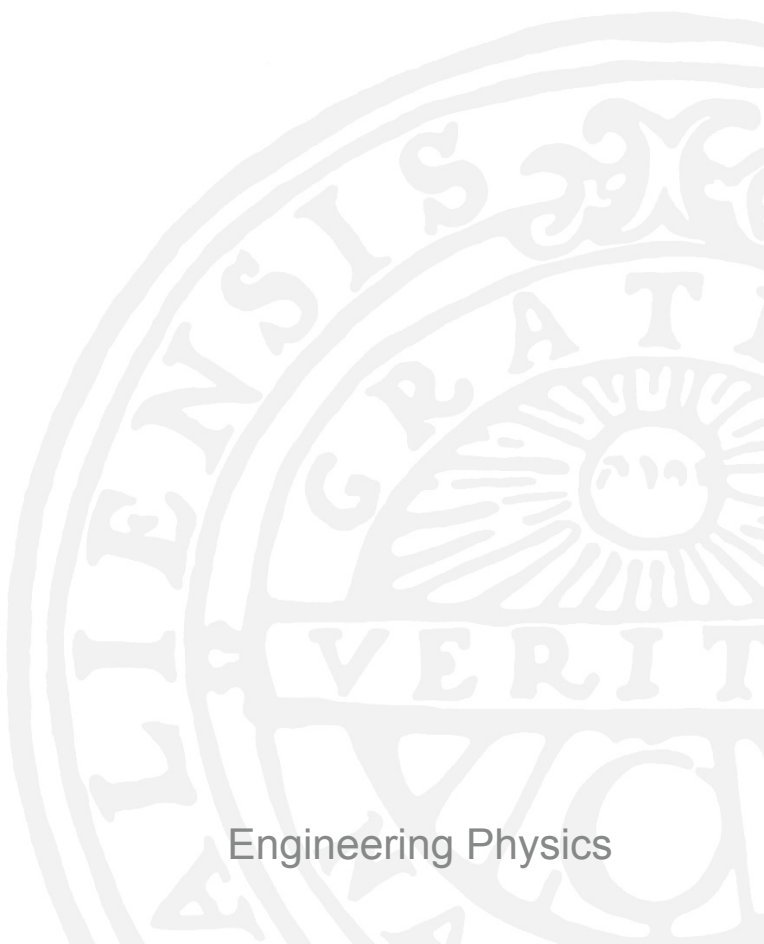
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A Mechanistic–Statistical Framework for Forecasting the Effective Reproduction Number in Epidemic Modelling

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Engineering Physics



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Abstract

Write the abstract here. Keep it within half a page and write it in English. Summarise the project background, aim, method, main results and conclusions concisely.

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Populärvetenskaplig sammanfattning

Skriv den populärvetenskapliga sammanfattningen här. Den ska vara ungefär 1–2 sidor lång och riktad till en läsare med kunskaper motsvarande gymnasienivå. Förklara bakgrund, metod, resultat och varför arbetet är relevant utan onödig fackterminologi.

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1 Introduction

1.1 Motivation

Epidemic forecasting aims to describe how disease transmission changes over time and to provide short-term predictions of how an outbreak may develop. A central quantity in this setting is the effective reproduction number, R_t , which reflects the level of transmission at a given time. Since R_t is not directly observable, it must be inferred from epidemic data.

This creates an important difficulty for forecasting. Mechanistic compartment models provide a structured description of disease spread, but their predictive performance depends on how well the transmission process can be recovered from available observations. In practice, reported epidemic data are often noisy and affected by delays or other sources of variability. As a result, accurate estimation of R_t remains challenging.

At the same time, statistical time-series models can capture temporal patterns directly from data, but they do not explicitly represent the disease dynamics that generate those patterns. Using either approach in isolation therefore imposes clear limitations.

These limitations motivate a hybrid forecasting approach. By combining mechanistic epidemic models with statistical time-series inference, it becomes possible to retain information about the underlying disease dynamics while allowing the forecasting component to adapt to temporal variation in transmission. The purpose is therefore to examine how these two modelling perspectives can be used together in a consistent forecasting framework.

1.2 Objective

The objective of this project is to evaluate a hybrid framework for forecasting the effective reproduction number, R_t . More specifically, the study aims to assess how the framework behaves under controlled synthetic conditions, how robust it remains when the modelling assumptions are relaxed, and how it performs when applied to real epidemiological data.

By considering the framework under these different conditions, the project aims to provide a more complete assessment of its predictive behaviour than would be possible in a single setting.

1.3 Study Design

The study is organised in three stages. In the first stage, analytically defined synthetic R_t trajectories are used to establish a controlled baseline for forecasting performance. Because the underlying transmission signal is known exactly in this setting, forecast accuracy can be assessed directly without uncertainty from data estimation or reporting variability.

In the second stage, the framework is examined under more challenging conditions by introducing model misspecification and more complex epidemic dynamics. This makes it possible to evaluate how sensitive the forecasting approach is when the assumptions of the baseline setting are no longer fully satisfied.

In the third stage, the framework is applied to real epidemiological data in order to assess its practical behaviour when the true transmission process is not directly known. Taken together, these three stages provide a progressive evaluation of the hybrid approach, from controlled validation to robustness analysis and real-world application.

2 Background and Theory

This chapter introduces the main concepts used in the forecasting framework. Epidemic compartment models and the effective reproduction number are presented first, since they define the dynamical setting of the problem. Statistical time-series models are then introduced as forecasting tools, followed by probabilistic forecast evaluation through the Weighted Interval Score. The chapter concludes with the hybrid perspective that combines mechanistic and statistical components within a common framework.

2.1 Epidemic Compartment Models and the Effective Reproduction Number

The SIRS compartment model

Compartment models describe epidemic dynamics by dividing a population into a set of epidemiological states and specifying how individuals move between them over time. In the SIR model, the population is divided into Susceptible (S), Infectious (I), and Recovered (R). The SIRS model extends this structure by allowing immunity to wane, so that recovered individuals eventually return to the susceptible class.

In its deterministic form, the SIRS model is given by

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta(t) S I}{N} + \xi R, \\ \frac{dI}{dt} &= \frac{\beta(t) S I}{N} - \gamma I, \\ \frac{dR}{dt} &= \gamma I - \xi R,\end{aligned}\tag{1}$$

where N is the total population size, $\beta(t)$ is the transmission rate, γ is the recovery rate, and ξ is the rate of loss of immunity. The total population is conserved, so that $S(t) + I(t) + R(t) = N$ for all t .

This model provides a simple but useful description of transmission dynamics. In particular, it gives a natural way to connect epidemic states to changes in transmission over time.

The effective reproduction number

A central quantity in epidemic forecasting is the effective reproduction number, R_t . It represents the expected number of secondary infections generated by a typical infectious individual at time t under the current transmission conditions.

The quantity R_t differs from the basic reproduction number R_0 , which refers to a fully susceptible population. By contrast, R_t reflects the transmission level during the course of an epidemic. When $R_t > 1$, transmission increases. When $R_t < 1$, transmission declines.

In the present setting, R_t is related to the transmission rate through

$$R_t = \frac{\beta(t)}{\gamma}.\tag{2}$$

Using this relation, the transmission term in (1) can be written in terms of R_t rather than $\beta(t)$.

In practice, R_t is not directly observable. It must be inferred from epidemic data, which are often noisy and affected by delays or other reporting effects. This is one of the main reasons why forecasting R_t is difficult. In the framework considered here, R_t is the quantity of interest and is treated as the forecasting target throughout the study.

2.2 Statistical Time-Series Models

Statistical time-series models describe how a variable evolves over time based on its past behaviour, and in some cases also on additional explanatory inputs. In this project, such models are used to forecast the temporal evolution of R_t .

Autoregressive models

An autoregressive model of order p , denoted $AR(p)$, expresses the current value of a time series as a linear combination of its previous values. For a series $y(t)$, the model is

$$y(t) = \sum_{i=1}^p a_i y(t-i) + \varepsilon(t), \quad (3)$$

where the coefficients a_i determine the dependence on past observations and $\varepsilon(t)$ is a stochastic error term.

The AR model is the simplest statistical model considered in the forecasting framework. Because it uses only past values of the target series, it serves as a natural baseline against which more elaborate models can be compared.

Autoregressive models with exogenous inputs

The ARX model extends the autoregressive structure by including external covariates. In the present context, these covariates may be mechanistic state variables such as the susceptible or infected population obtained from a compartment model.

A general ARX model can be written as

$$y(t) = \sum_{i=1}^{n_a} a_i y(t-i) + \sum_{j=0}^{n_b-1} \mathbf{b}_j^\top \mathbf{u}(t-n_k-j) + \varepsilon(t), \quad (4)$$

where $\mathbf{u}(t)$ denotes the exogenous input, n_a is the autoregressive order, n_b is the number of input lags, and n_k is the input delay.

The main idea behind ARX is that the time series may contain useful information not only in its own past values, but also in external variables that are related to the underlying dynamics. This is particularly relevant in hybrid forecasting, where mechanistic model states can be used to support statistical prediction.

Discrete-time linear state-space models

State-space models provide a more general description of temporal dynamics by introducing latent state variables. A linear discrete-time state-space model has the form

$$\begin{aligned} \mathbf{x}(t+1) &= A \mathbf{x}(t) + B \mathbf{u}(t) + \mathbf{w}(t), \\ y(t) &= C \mathbf{x}(t) + D \mathbf{u}(t) + v(t), \end{aligned} \quad (5)$$

where $\mathbf{x}(t)$ is the latent state vector, $\mathbf{u}(t)$ is an optional exogenous input, and $y(t)$ is the observed output. The terms $\mathbf{w}(t)$ and $v(t)$ represent process and observation noise.

This formulation is more flexible than a standard AR or ARX model because it allows the observed time series to be driven by hidden internal states. In the broader forecasting framework, state-space identification methods such as N4SID and SSEST are considered as extensions of the simpler autoregressive models.

2.3 Probabilistic Forecasts and the Weighted Interval Score

Probabilistic forecasts

A probabilistic forecast does not provide only a single predicted value. Instead, it describes the uncertainty around the prediction through a distribution or through summary quantities derived from that distribution, such as predictive intervals.

For epidemic forecasting, this is important because uncertainty is an inherent part of the problem. A forecast should therefore not only indicate the most likely future trajectory, but also how certain or uncertain that prediction is.

A central predictive interval at nominal coverage level $1 - \alpha$ is defined by a lower bound ℓ_α and an upper bound u_α . If the forecast is well calibrated, the observed value should fall within this interval approximately $(1 - \alpha) \times 100\%$ of the time.

Weighted Interval Score

This study uses the Weighted Interval Score (WIS) to evaluate probabilistic forecasts represented by predictive intervals. WIS is a proper scoring rule and has been used in epidemic forecast evaluation, including during COVID-19^[1].

For an observed value y , a predictive median m , and K central prediction intervals with miscoverage rates α_k , lower bounds ℓ_k , and upper bounds u_k , the WIS is defined as

$$\text{WIS}(y) = \frac{1}{K + \frac{1}{2}} \left[\frac{1}{2} |y - m| + \sum_{k=1}^K \frac{\alpha_k}{2} \text{IS}_{\alpha_k}(y) \right], \quad (6)$$

where the interval score is

$$\text{IS}_{\alpha_k}(y) = (u_k - \ell_k) + \frac{2}{\alpha_k} \max(\ell_k - y, 0) + \frac{2}{\alpha_k} \max(y - u_k, 0). \quad (7)$$

The first term in the interval score measures the width of the predictive interval. The two remaining terms penalise cases where the observed value falls below or above the interval. This means that WIS reflects both sharpness and calibration. Narrow intervals are preferred, but only if they still capture the observed value at an appropriate rate.

Because the forecasting framework considered here produces probabilistic forecasts rather than only point predictions, WIS provides a more informative evaluation measure than metrics based solely on pointwise error.

2.4 Hybrid Mechanistic–Statistical Forecasting

Mechanistic and statistical forecasting approaches have complementary strengths. Mechanistic models provide a structured description of disease transmission and are directly connected to epidemiological assumptions. Statistical models are more flexible and can adapt to patterns in observed data without requiring the full epidemic process to be specified explicitly.

A hybrid framework aims to combine these two perspectives. One common approach is to use a mechanistic model to generate internal state variables or other dynamical summaries, and then use these quantities as inputs to a statistical forecasting model. This allows the statistical component to exploit mechanistic structure without requiring the entire forecasting problem to be handled within the mechanistic model alone.

The framework considered in this project follows this general idea. Compartment-model states are used as exogenous information for statistical forecasting models, while the autoregressive baseline is retained for comparison. This makes it possible to assess whether mechanistic state

information improves forecasts of R_t relative to models that rely only on the history of the target series itself.

This hybrid perspective provides the theoretical basis for the methodological choices introduced in the next chapter.

3 Method

This chapter describes how the forecasting framework is evaluated under a controlled synthetic validation setting. The analysis begins with a set of predefined transmission scenarios for R_t , followed by deterministic generation of the corresponding SIRS state trajectories. Statistical forecasting models are then applied under several model and covariate configurations. Hyperparameters are selected globally, after which forecast performance is evaluated using an expanding-window procedure and the Weighted Interval Score.

3.1 Overview

The methodological workflow consists of four main components. First, synthetic R_t trajectories are constructed to represent different transmission patterns. Second, these trajectories are used to generate consistent mechanistic state variables from the SIRS model. Third, statistical forecasting models are fitted under a range of configurations and hyperparameters. Fourth, the selected models are evaluated through expanding-window forecasting using probabilistic performance measures.

3.2 Synthetic Scenarios

To evaluate the forecasting framework under controlled conditions, four synthetic R_t scenarios were defined using analytic functions. Each trajectory spans 365 days and represents a distinct transmission pattern while remaining within realistic bounds. These predefined trajectories serve as the ground-truth transmission signals in the synthetic validation setting considered here.

The four transmission patterns are:

- A1 — Seasonal Forcing.** A sinusoidal R_t profile that produces periodic epidemic waves over time.
- A2 — Policy Intervention.** A sigmoid-based transition representing a gradual reduction in transmission, analogous to the introduction of a public-health intervention.
- A3 — Damping Resurgence.** A multi-wave Gaussian construction in which successive peaks decrease in magnitude, representing a gradually weakening outbreak.
- A4 — Amplifying Resurgence.** A multi-wave Gaussian construction in which successive peaks increase in magnitude, representing a progressively intensifying outbreak.

Because the true evolution of each transmission trajectory is known exactly, forecast performance can be assessed directly against the underlying signal without uncertainty from observation noise or data estimation. These predefined scenarios therefore provide a controlled basis for the forecasting analysis developed in the following sections.

3.3 Deterministic SIRS State Generation

For each predefined R_t trajectory, corresponding mechanistic state variables were generated using the deterministic SIRS model described in Section 2.1. The model was parameterised with a population size $N = 100,000$, a recovery rate $\gamma = 1/7 \text{ day}^{-1}$, and an immunity-loss rate $\xi = 1/90 \text{ day}^{-1}$. The initial number of infected individuals was set to $I_0 = 500$ for scenarios A1, A3, and A4, and to $I_0 = 5,000$ for scenario A2.

The deterministic solver was used to produce Susceptible and Infected trajectories that are consistent with the prescribed R_t signal. These state variables were later used as exogenous inputs in the models that incorporate mechanistic information. Since the solver is deterministic, the generated state trajectories remain free from stochastic variability. This allows the forecasting component to be evaluated without introducing additional noise through the mechanistic states.

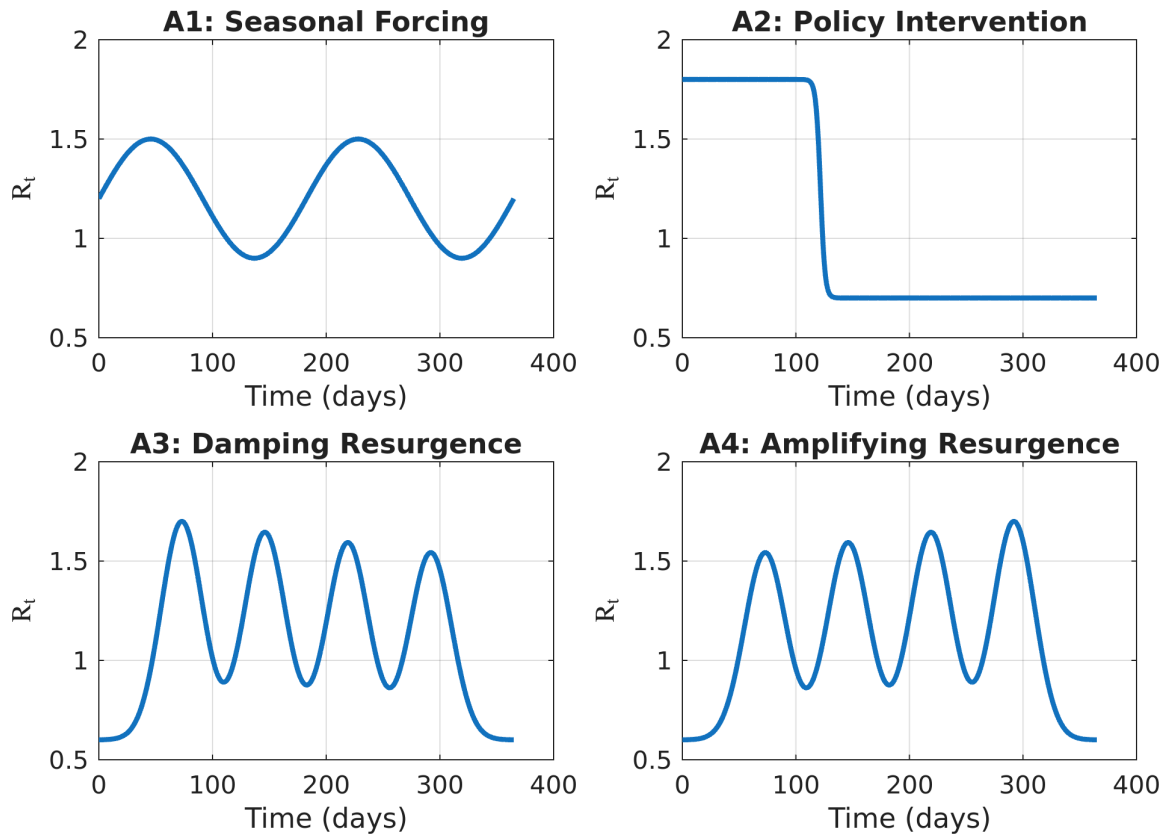


Figure 1: Synthetic R_t trajectories used in the synthetic validation setting. The four scenarios represent seasonal forcing, policy intervention, damping resurgence, and amplifying resurgence, and serve as the predefined transmission signals for the forecasting analysis.

Both $S(t)$ and $I(t)$ were scaled by the population size N before being used as model inputs.

Projection of future exogenous inputs

When models with exogenous inputs are used, forecasts over the prediction horizon require future values of the mechanistic covariates. These values are not observed directly and must therefore be projected.

To obtain them, the deterministic SIRS solver was run forward from the current window endpoint under a flat- β assumption, meaning that the transmission rate was held constant over the forecast horizon at its most recent observed value. This provides a consistent short-term projection of the S and I trajectories without assuming knowledge of future R_t .

3.4 Forecasting Models and Configurations

Four families of statistical time-series models were considered: AR, ARX, N4SID, and SSEST. Their theoretical forms were introduced in Section 2.2; here the focus is on how they are used in the synthetic validation analysis.

All models were fitted to the log-transformed transmission series,

$$y(t) = \log R_t(t),$$

in order to improve numerical stability. Forecasts were then transformed back to the natural scale for evaluation.

AR

The AR model serves as the statistical baseline and uses only past values of the log-transformed R_t series. The autoregressive order was searched over $p \in \{0, 1, \dots, 14\}$.

ARX

The ARX model extends the AR structure by including exogenous covariates. In this study, these covariates are mechanistic state variables derived from the SIRS model. The hyperparameter grid consists of the autoregressive order $n_a \in \{0, \dots, 14\}$, the exogenous memory order $n_b \in \{1, \dots, 7\}$, and the input delay $n_k \in \{1, \dots, 7\}$.

N4SID and SSEST

N4SID and SSEST are discrete-time linear state-space identification methods. Both were evaluated with model order $n \in \{1, \dots, 8\}$. These methods extend the simpler autoregressive setting by allowing the observed series to depend on latent states, with or without mechanistic covariates.

Covariate configurations

ARX, N4SID, and SSEST were evaluated under three mechanistic input configurations: Susceptible only (*S*), Infected only (*I*), and both compartments jointly (*Both*). In addition, AR operates without exogenous inputs, and N4SID and SSEST were also evaluated in a *None* configuration.

Table 1: Model families and covariate configurations used in the synthetic validation setting.

Model family	Configurations
AR	None
ARX	S, I, Both
N4SID	None, S, I, Both
SSEST	None, S, I, Both

This gives 12 model–configuration combinations in total.

Probabilistic forecasts

All model families produce probabilistic forecasts rather than only point predictions. Each forecast consists of a predictive median together with four symmetric predictive intervals at miscoverage levels

$$\alpha \in \{0.05, 0.10, 0.20, 0.50\},$$

corresponding to 95%, 90%, 80%, and 50% predictive intervals. The interval-based representation allows forecast uncertainty to be evaluated explicitly using the Weighted Interval Score introduced in Section 2.3.

Fallback rule

When model estimation failed, produced non-finite values, or returned an almost constant signal in log-space, a persistence forecast was used at the model-fitting stage. In this case, the forecast was defined as a constant continuation of the last observed R_t value. Forecast outputs that remained invalid after this step were penalised during hyperparameter selection and evaluation in order to avoid selecting numerically degenerate configurations.

3.5 Global Hyperparameter Selection

Hyperparameter selection was carried out before the final evaluation phase. For each model family and covariate configuration, a single hyperparameter setting was selected globally rather than separately for each forecast window.

The selection procedure was based on the Weighted Interval Score. For every candidate configuration within a given model family, expanding-window forecasts were generated across all four scenarios. The mean WIS was then computed within each scenario, and these scenario-level means were averaged equally to obtain a single global score for the candidate. The configuration with the lowest global mean WIS was selected and fixed for subsequent forecasting.

This design ensures that hyperparameter selection is based on performance across the full set of synthetic transmission patterns rather than on a single scenario or window.

3.6 Expanding-Window Forecasting

Forecast performance was evaluated using an expanding-window procedure that reflects sequential data availability. Each forecast began with an initial training window of 49 days of historical R_t values. Based on this window, the model parameters were estimated and a 14-day forecast was generated.

After each forecast, the window endpoint was advanced by 7 days. The model was then re-estimated using the expanded set of observations, and a new 14-day forecast was produced. This process was repeated throughout the 365-day timeline until the remaining data length equalled the forecast horizon. In total, this yielded 44 forecast windows per scenario.

For models with exogenous inputs, future mechanistic covariates over the forecast horizon were supplied using the projected SIRS trajectories described in Section 3.3.

3.7 Forecast Evaluation

Forecast performance was quantified using the Weighted Interval Score (WIS), which is defined in Section 2.3. Since the models produce probabilistic forecasts, WIS provides a more informative measure than pointwise error metrics alone.

For each forecast window, WIS was computed at each of the 14 forecast steps and then averaged across the horizon to obtain a per-window score. These scores were subsequently averaged across all windows within a scenario to obtain a scenario-level mean WIS. Comparison across model families and configurations was then based on these aggregated scores.

This evaluation design makes it possible to compare forecast behaviour across scenarios, model structures, and exogenous-input settings under a common probabilistic criterion.

4 Results and Discussion

This chapter presents the forecasting results obtained under the synthetic validation setting and discusses the main patterns that emerge across scenarios, model families, and covariate configurations. The results are reported using the Weighted Interval Score (WIS), which evaluates the full probabilistic forecast rather than only pointwise error.

4.1 Overall Forecast Performance

Forecasts were evaluated over 44 expanding windows per scenario, each with a 14-day forecast horizon. Starting from a minimum training window of 49 days and advancing in steps of 7 days, this yielded a total of 29,568 scored forecast steps across the twelve model–configuration combinations considered here.

Hyperparameters were selected globally before the final evaluation. For each model family and covariate configuration, a single setting was chosen by minimising the mean WIS across all four scenarios. The selected configurations were then held fixed throughout the expanding-window evaluation.

Figure 2 summarises the window-level WIS distributions across all model families and scenarios. The ARX variants achieve the lowest and most consistent scores overall, especially on the smoother transmission patterns. By contrast, the state-space models show substantially larger dispersion, and scenario A2 stands out as the most difficult case across all model families.

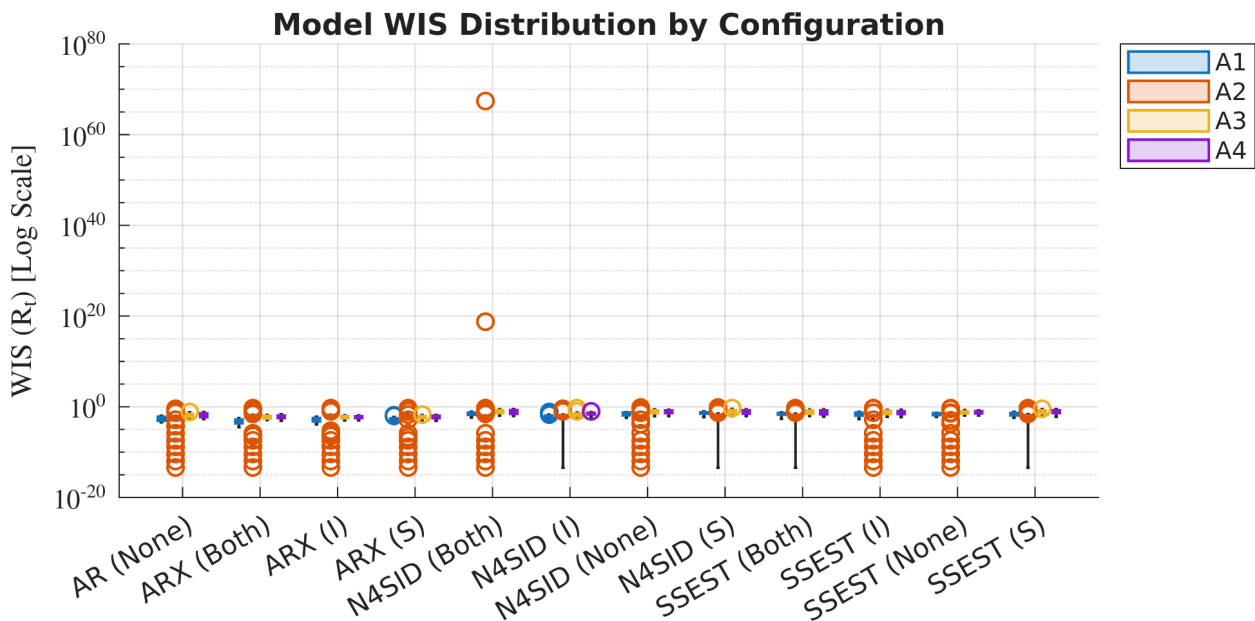


Figure 2: Window-level WIS distributions for all model–configuration combinations across the four synthetic scenarios. Colours indicate scenarios A1–A4. Only finite WIS values are included in the plot. Since the scores span several orders of magnitude, the vertical axis is displayed on a logarithmic scale.

Table 2 shows the mean window WIS for each combination and scenario, together with the global mean obtained by averaging the four scenario means equally. The three ARX variants occupy the top three global ranks. ARX/I achieves the lowest global mean WIS, followed closely by ARX/S and ARX/Both. The autoregressive baseline ranks fourth, while the state-space models perform less favourably overall.

Taken together, the overall ranking indicates that models using mechanistic covariates through the ARX structure provide the strongest probabilistic forecasts in the synthetic setting considered

Table 2: Mean window WIS per model–configuration combination and scenario. Lower values indicate better probabilistic forecast performance. The global mean is the unweighted average of the four scenario means. [†] N4SID/Both is excluded from the global ranking because of numerical blow-up on scenario A2.

Model	Exo	A1	A2	A3	A4	Global mean
ARX	I	2.174×10^{-3}	4.295×10^{-2}	5.693×10^{-3}	5.894×10^{-3}	1.418×10^{-2}
ARX	S	2.015×10^{-3}	4.256×10^{-2}	6.364×10^{-3}	6.252×10^{-3}	1.430×10^{-2}
ARX	Both	9.545×10^{-4}	4.377×10^{-2}	6.468×10^{-3}	6.841×10^{-3}	1.451×10^{-2}
AR	None	3.961×10^{-3}	2.442×10^{-2}	2.012×10^{-2}	2.123×10^{-2}	1.743×10^{-2}
N4SID	I	1.129×10^{-2}	3.166×10^{-2}	3.194×10^{-2}	3.422×10^{-2}	2.728×10^{-2}
SSEST	None	2.632×10^{-2}	3.410×10^{-2}	6.794×10^{-2}	7.114×10^{-2}	4.988×10^{-2}
SSEST	I	3.330×10^{-2}	2.961×10^{-2}	8.225×10^{-2}	7.337×10^{-2}	5.463×10^{-2}
SSEST	Both	3.113×10^{-2}	3.528×10^{-2}	9.493×10^{-2}	9.006×10^{-2}	6.285×10^{-2}
N4SID	None	3.491×10^{-2}	3.333×10^{-2}	1.020×10^{-1}	1.057×10^{-1}	6.899×10^{-2}
SSEST	S	2.948×10^{-2}	3.550×10^{-2}	1.189×10^{-1}	1.135×10^{-1}	7.434×10^{-2}
N4SID	S	4.468×10^{-2}	4.371×10^{-2}	1.441×10^{-1}	1.027×10^{-1}	8.378×10^{-2}
N4SID	Both [†]	3.503×10^{-2}	5.11×10^{65}	1.128×10^{-1}	1.032×10^{-1}	—

here. The baseline AR model remains competitive, particularly when the underlying dynamics are abrupt. The state-space models show more variable behaviour and appear to be more sensitive to the choice of covariates and scenario structure.

4.2 Scenario-Dependent Forecast Behaviour

Forecast behaviour depends strongly on the transmission trajectory. Smooth and gradually varying scenarios are forecasted more accurately than scenarios with abrupt structural changes.

Scenario A1 provides a representative example of the smooth case. In this setting, the ARX models track the oscillatory pattern closely across successive windows. Forecast performance is stable, and the differences between the three ARX configurations remain small. Figures 3 and 4 illustrate this behaviour for ARX/I and ARX/S respectively.

The multi-wave scenarios A3 and A4 follow the same general pattern. ARX variants achieve consistently low WIS values, whereas the state-space models show substantially higher errors. For example, ARX/I records mean WIS values of 5.693×10^{-3} on A3 and 5.894×10^{-3} on A4, while N4SID/None reaches 1.020×10^{-1} and 1.057×10^{-1} on the same scenarios. This indicates that the autoregressive structure with mechanistic covariates is able to adapt well to repeated but smooth transmission changes, whereas the state-space estimators struggle more with oscillatory dynamics.

Scenario A2 is qualitatively different. Here the transmission signal contains a sharp policy-like transition, and this produces the most challenging forecasting setting in the study. In contrast to the smoother scenarios, AR/None achieves the lowest mean WIS on A2, with a value of 2.442×10^{-2} . Figure 5 illustrates this case.

The main difficulty in A2 is that forecasts issued before the transition cannot anticipate the new transmission level. Since the models are trained on data generated under the previous regime, their forecasts remain close to that earlier level and only adjust once enough post-transition observations have entered the expanding window. This produces a systematic lag around the structural break.

A2 also shows high variability in window-level WIS. For example, AR/None has a mean window WIS of 2.442×10^{-2} but a median of 7.765×10^{-3} . This indicates that the elevated error is driven mainly by a smaller number of difficult windows around the transition itself, while most windows away from the structural break are forecasted much more accurately.

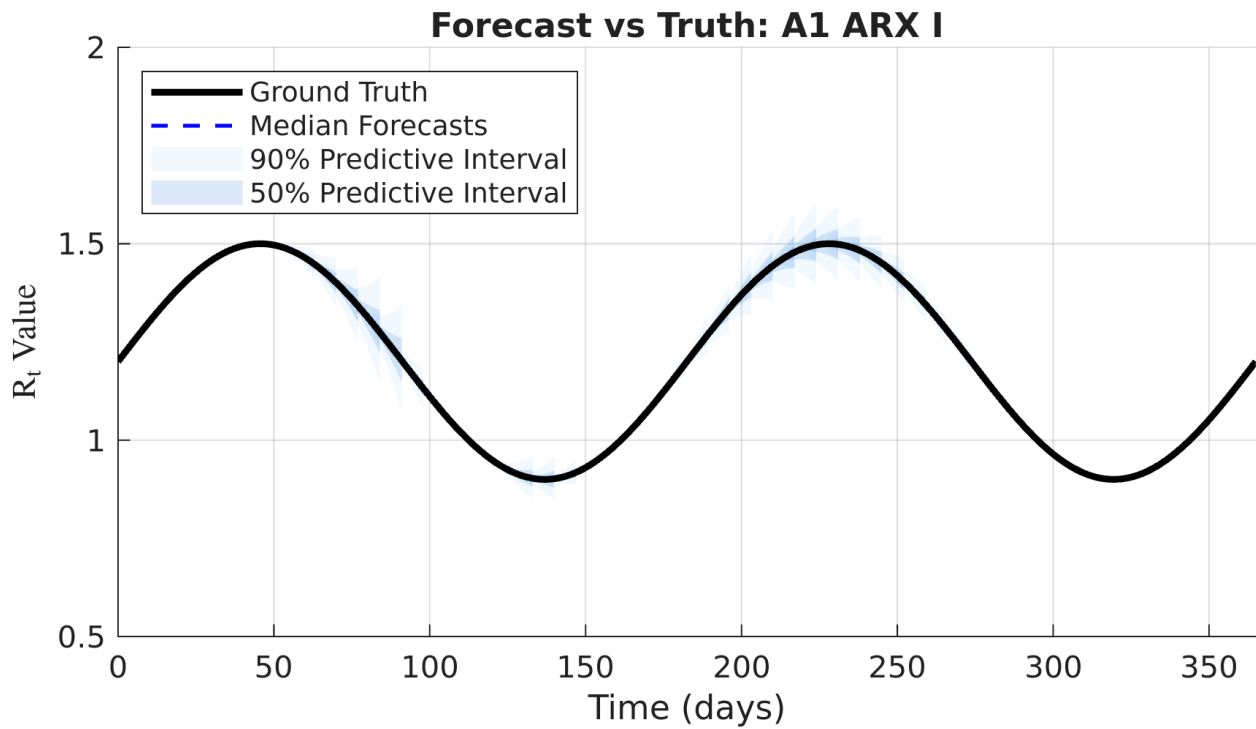


Figure 3: ARX/I expanding-window forecasts on scenario A1 (seasonal forcing). Each band represents one 14-day forecast window, while the black line shows the true R_t trajectory. The mean window WIS on this scenario is 2.174×10^{-3} .

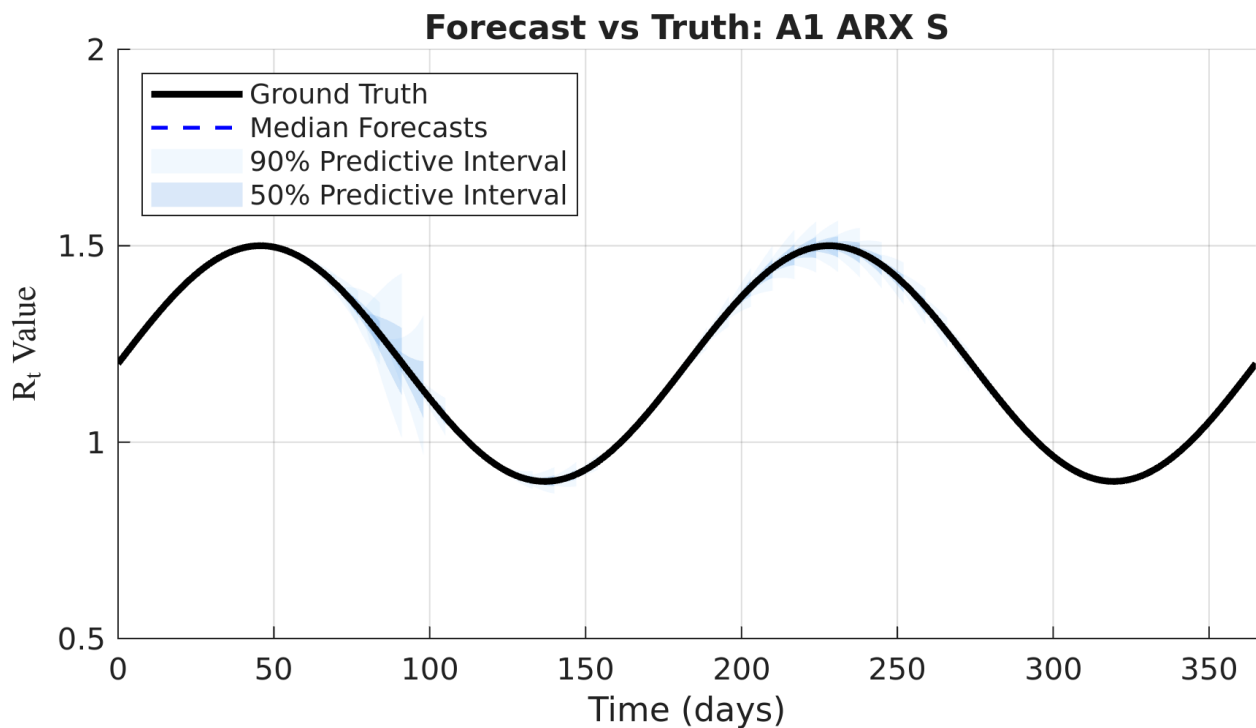


Figure 4: ARX/S expanding-window forecasts on scenario A1 (seasonal forcing). The mean window WIS on this scenario is 2.015×10^{-3} , indicating performance similar to that of ARX/I.

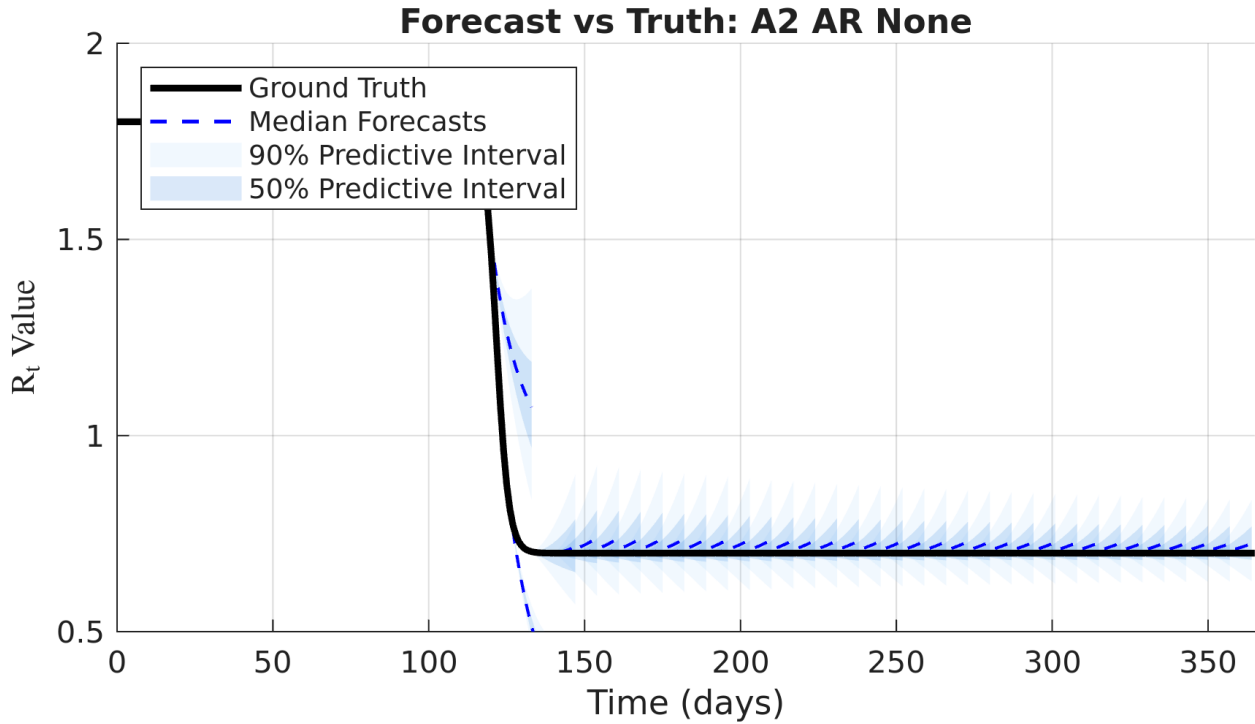


Figure 5: AR/None expanding-window forecasts on scenario A2 (policy intervention). The mean window WIS is 2.442×10^{-2} , which is the lowest value among all model configurations on this scenario.

Additional forecast plots for the remaining combinations are provided in Appendix A.

4.3 Role of Exogenous Inputs

The effect of mechanistic covariates depends on both the model family and the transmission scenario.

Within the ARX family, exogenous inputs are clearly beneficial on the smooth and multi-wave scenarios. All three ARX configurations outperform AR/None on A1, A3, and A4. On A1, for instance, ARX/Both achieves a mean WIS of 9.545×10^{-4} , compared with 3.961×10^{-3} for AR/None. This corresponds to roughly a fourfold improvement.

The differences among the ARX variants themselves are relatively modest on the smooth scenarios. On A1, the spread across ARX/S, ARX/I, and ARX/Both ranges from 9.545×10^{-4} to 2.174×10^{-3} . This suggests that, under smooth transmission dynamics, the Susceptible and Infected compartments provide broadly similar useful information to the autoregressive framework.

The situation changes on scenario A2. Here the ordering is reversed, and the baseline AR model outperforms all ARX variants. AR/None reaches 2.442×10^{-2} , whereas the three ARX configurations range from 4.256×10^{-2} to 4.377×10^{-2} . In this setting, the mechanistic covariates are themselves strongly affected by the abrupt regime change. Rather than providing stabilising information, they appear to introduce additional non-stationary structure that is difficult for the ARX model to use productively.

For the state-space models, the Infected-compartment mode is the most effective exogenous configuration. N4SID/I achieves a global mean WIS of 2.728×10^{-2} , compared with 6.899×10^{-2} for N4SID/None. By contrast, N4SID/S performs worse than the None configuration. A similar pattern appears for SSEST, where SSEST/None outperforms SSEST/S, while SSEST/I lies between them.

One possible explanation is that the Susceptible compartment changes relatively slowly within the forecast horizon and therefore contributes limited short-term information about variation in R_t . The Infected compartment responds more directly to transmission changes and may therefore provide a more informative within-window signal for models that identify input–output relationships. This effect is especially visible for the state-space estimators, which appear more sensitive than ARX to the usefulness of the chosen covariates.

Taken together, these results show that exogenous mechanistic information can be valuable, but its benefit is not universal. It depends on both the forecast model and the type of transmission dynamics being considered.

4.4 Model Limitations and Failure Cases

The most striking failure case in the present experiments is N4SID/Both on scenario A2. In this case, the mean window WIS reaches 5.11×10^{65} , far beyond the range of all other results. Figure 6 illustrates this behaviour.

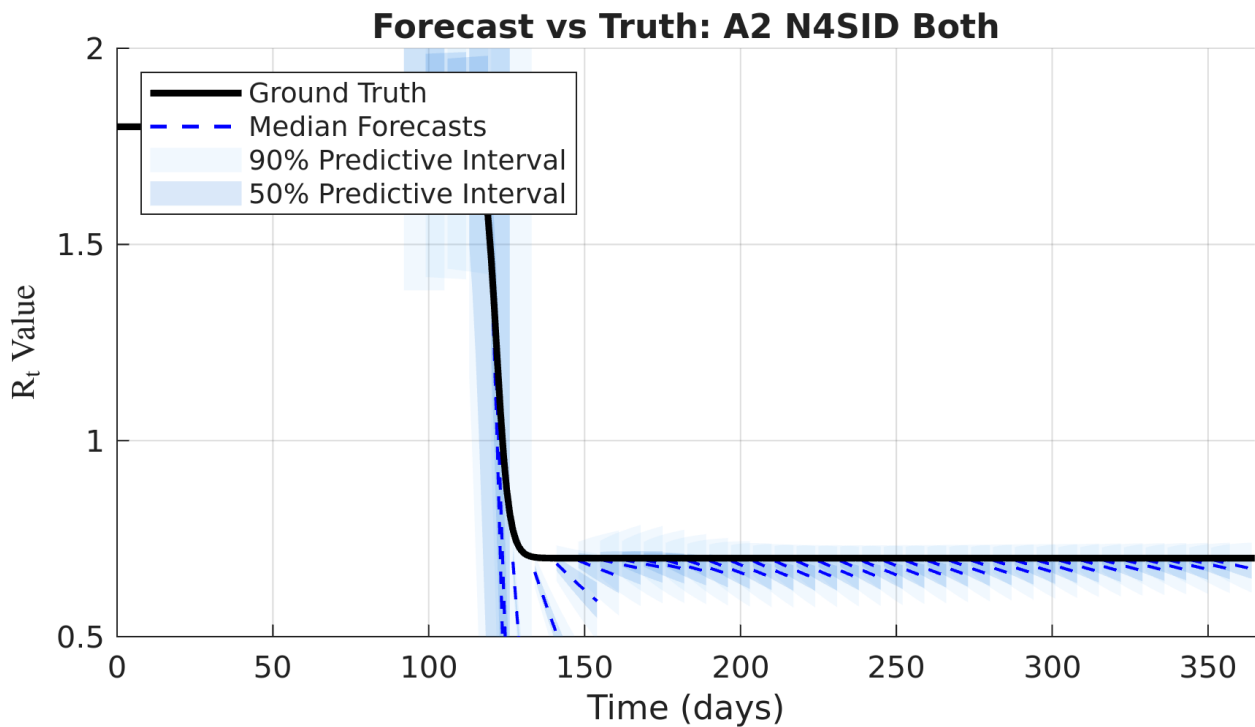


Figure 6: N4SID/Both expanding-window forecasts on scenario A2. The extreme divergence during the transition corresponds to a mean window WIS of 5.11×10^{65} .

This failure is specific to the combination of the abrupt policy-intervention scenario and the joint two-input exogenous mode. The same model configuration remains finite on A1, A3, and A4, and SSEST/Both does not show the same instability on A2. This suggests that the problem lies in the numerical behaviour of the N4SID identification step under rapidly changing multi-input conditions rather than in the state-space formulation itself.

A second limitation concerns the interpretation of the synthetic validation setting more generally. All results reported here are based on trajectories generated under the same SIRS structure that is later used to project exogenous covariates during forecasting. This means that models using mechanistic inputs operate under idealised conditions in which the projected covariates are structurally consistent with the generating process. In real-data applications, this advantage will be weaker, since future compartment states are not known and must themselves be approximated.

A further limitation concerns the persistence fallback. When model fitting fails or produces degenerate outputs, the framework substitutes a constant continuation of the most recent R_t value. This avoids numerical breakdown during evaluation, but it also means that some forecast failures may be absorbed into aggregate WIS values rather than appearing directly as separate events. The current analysis does not quantify how often this fallback was triggered.

Finally, scenario A2 is structurally distinct from the other three cases. It begins from a higher initial prevalence and contains an abrupt regime shift rather than a smooth or oscillatory transmission pattern. Conclusions drawn from A2 should therefore be understood as evidence about structural-break behaviour rather than as representative of all epidemic dynamics considered in the study.

Overall, the results indicate that the hybrid autoregressive framework performs strongly when transmission varies smoothly, but abrupt regime changes remain a major source of difficulty. They also show that the usefulness of mechanistic covariates depends on the forecast model and the transmission setting, and that state-space identification methods may be more sensitive to numerical instability under challenging conditions.

5 Conclusions

Write the conclusions here. Summarise the main findings of the study, relate them to the overall objective of the project, and discuss the broader implications of the results once the full report has been completed.

References

- [1] Johannes Bracher, Evan L. Ray, Tilmann Gneiting, and Nicholas G. Reich. Evaluating epidemic forecasts in an interval format. *PLOS Computational Biology*, 17(2):e1008618, February 2021. ISSN 1553-7358. doi: 10.1371/journal.pcbi.1008618.

A Supplementary Forecast Plots

This appendix collects expanding-window forecast plots that are not shown in the main text. The figures are grouped by model family in order to support the comparisons discussed in Section 4.

In each figure, the coloured bands represent successive 14-day forecast windows, including the predictive median together with the 50% and 90% predictive intervals. The black line shows the true R_t trajectory.

The N4SID/Both configuration on scenario A2 is not repeated here, since that failure case is already shown and discussed in Section 4.4.

AR baseline

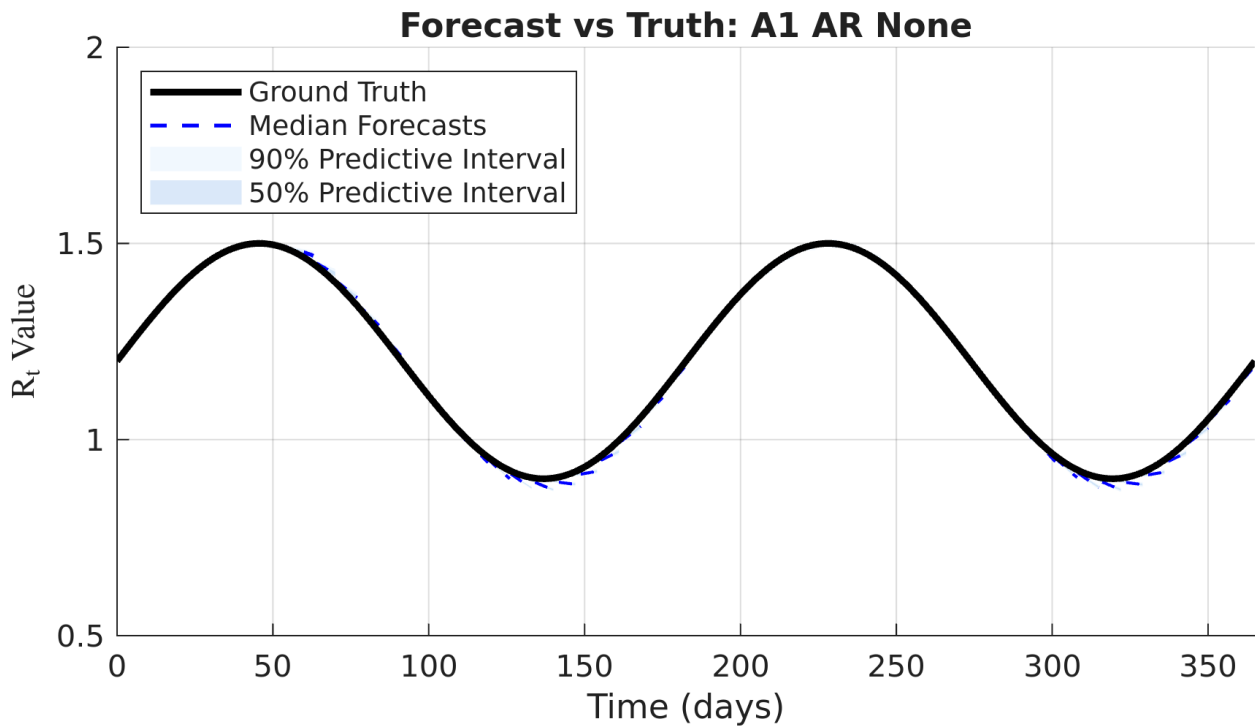


Figure 7: AR/None on scenario A1 (seasonal forcing). Mean window WIS = 3.96×10^{-3} .

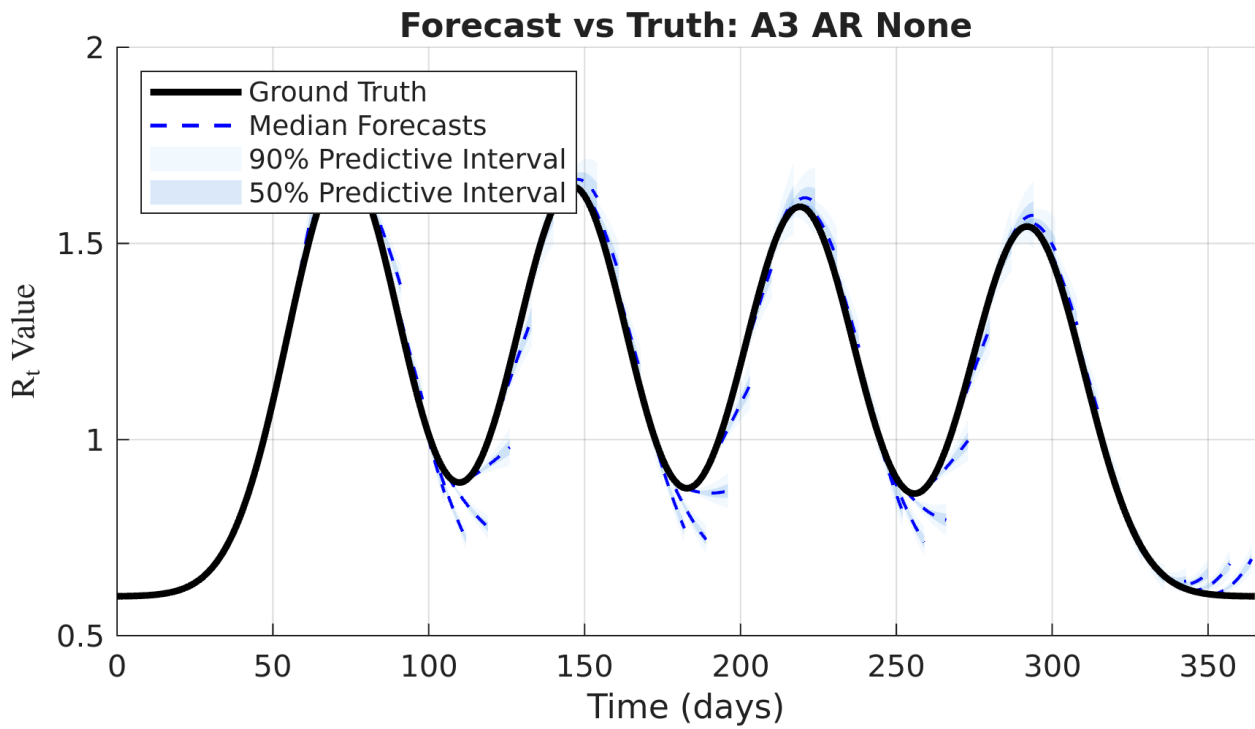


Figure 8: AR/None on scenario A3 (damping resurgence). Mean window WIS = 2.01×10^{-2} .

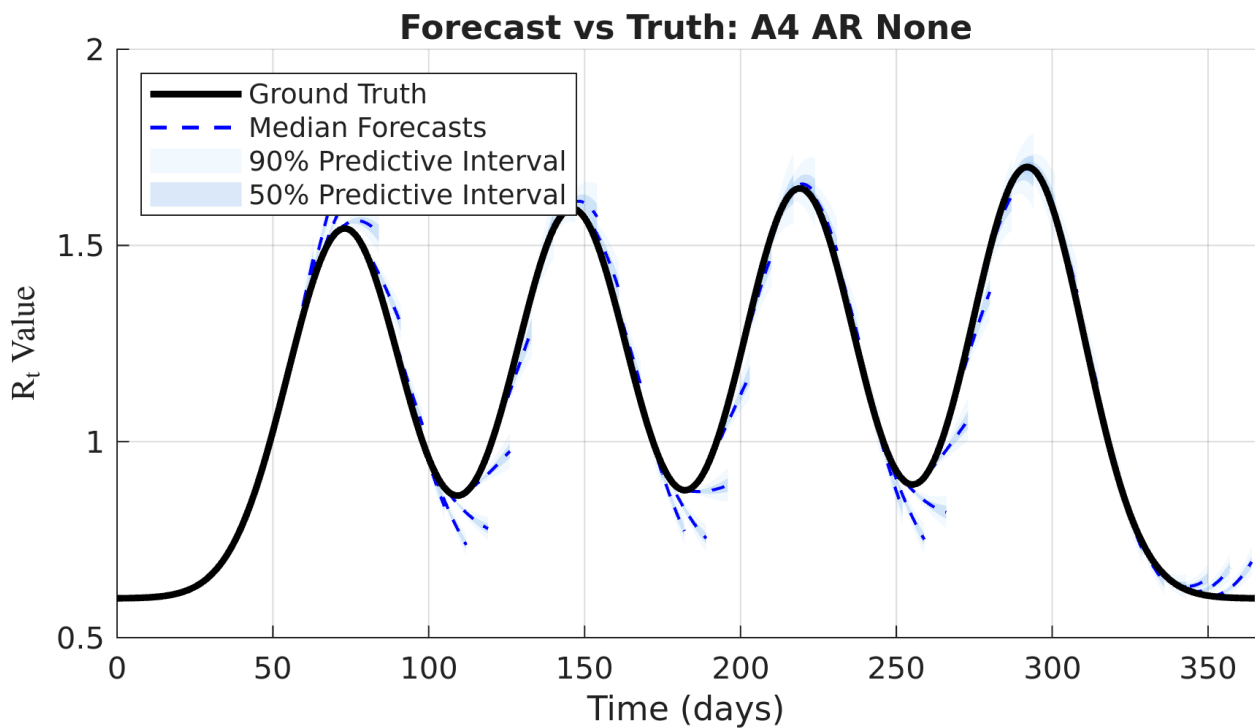


Figure 9: AR/None on scenario A4 (amplifying resurgence). Mean window WIS = 2.12×10^{-2} .

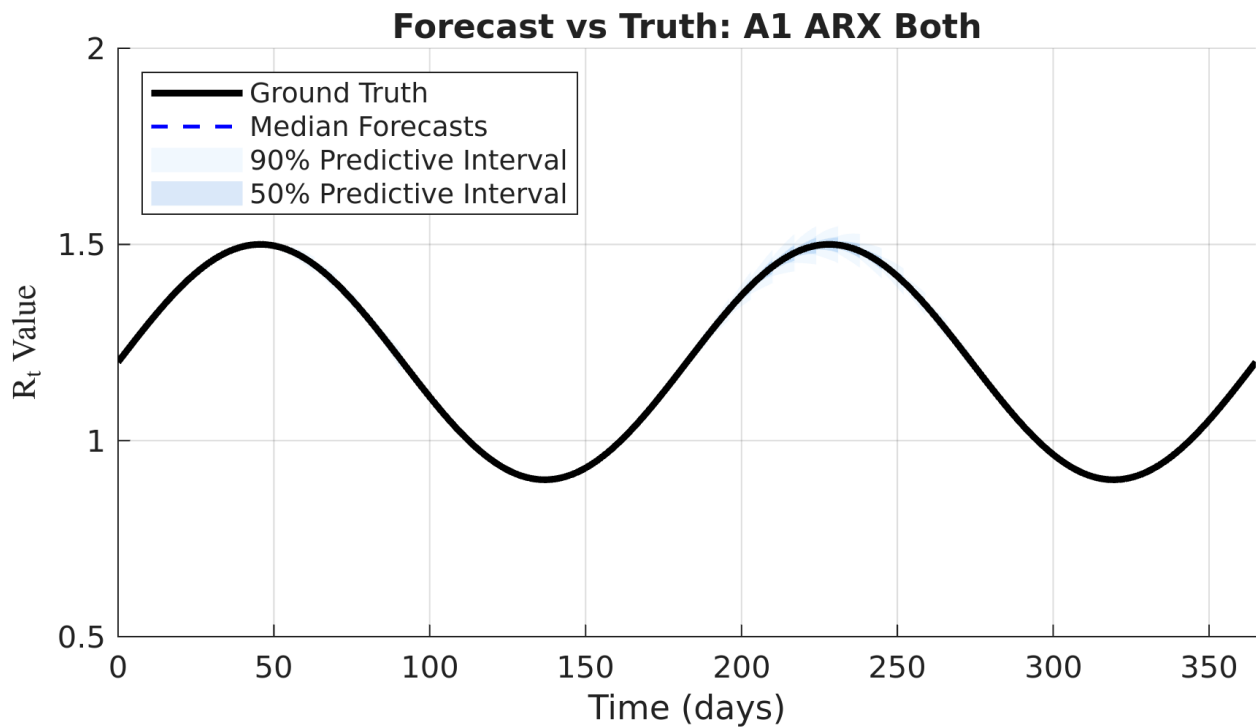
ARX

Figure 10: ARX/Both on scenario A1 (seasonal forcing). Mean window WIS = 9.54×10^{-4} .

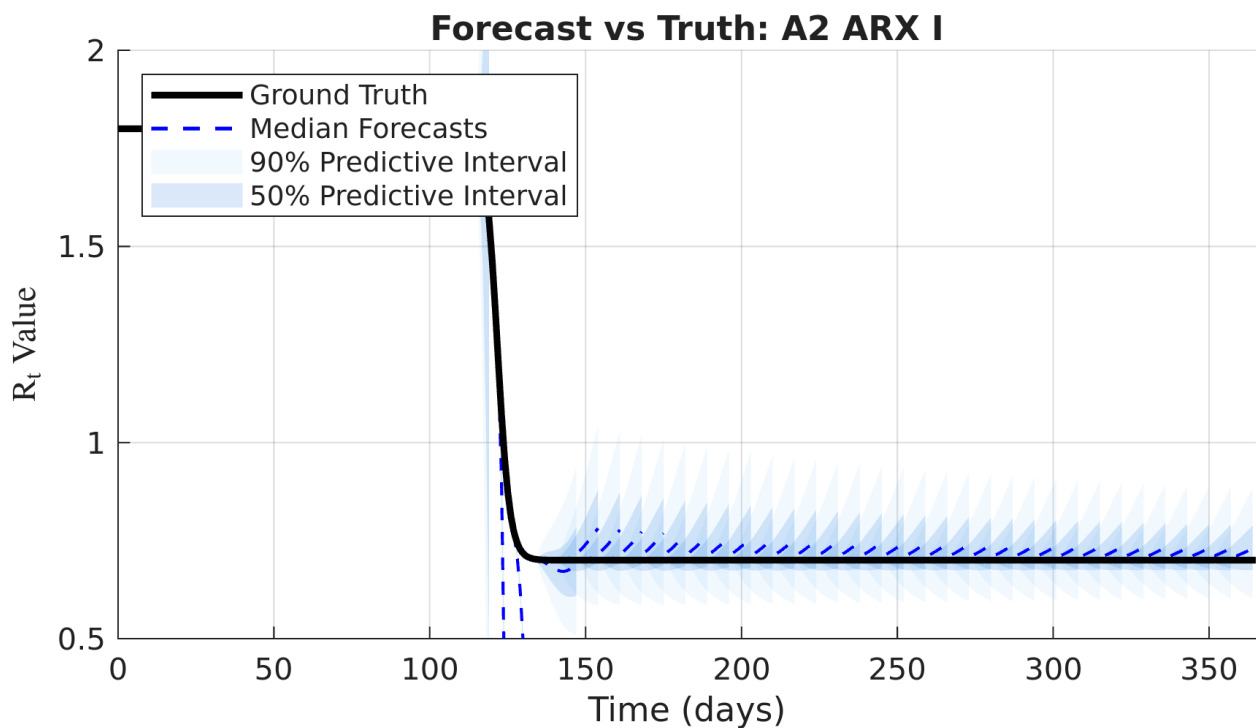


Figure 11: ARX/I on scenario A2 (policy intervention). Mean window WIS = 4.30×10^{-2} .

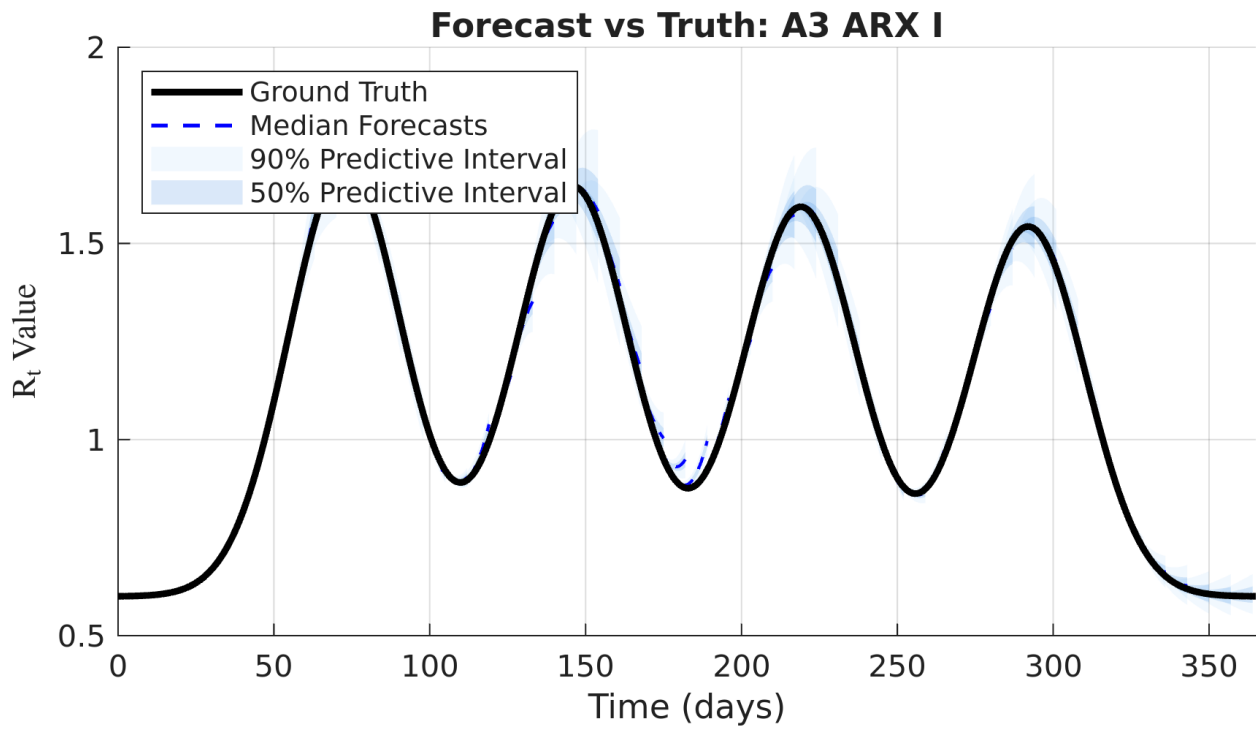


Figure 12: ARX/I on scenario A3 (damping resurgence). Mean window WIS = 5.69×10^{-3} .

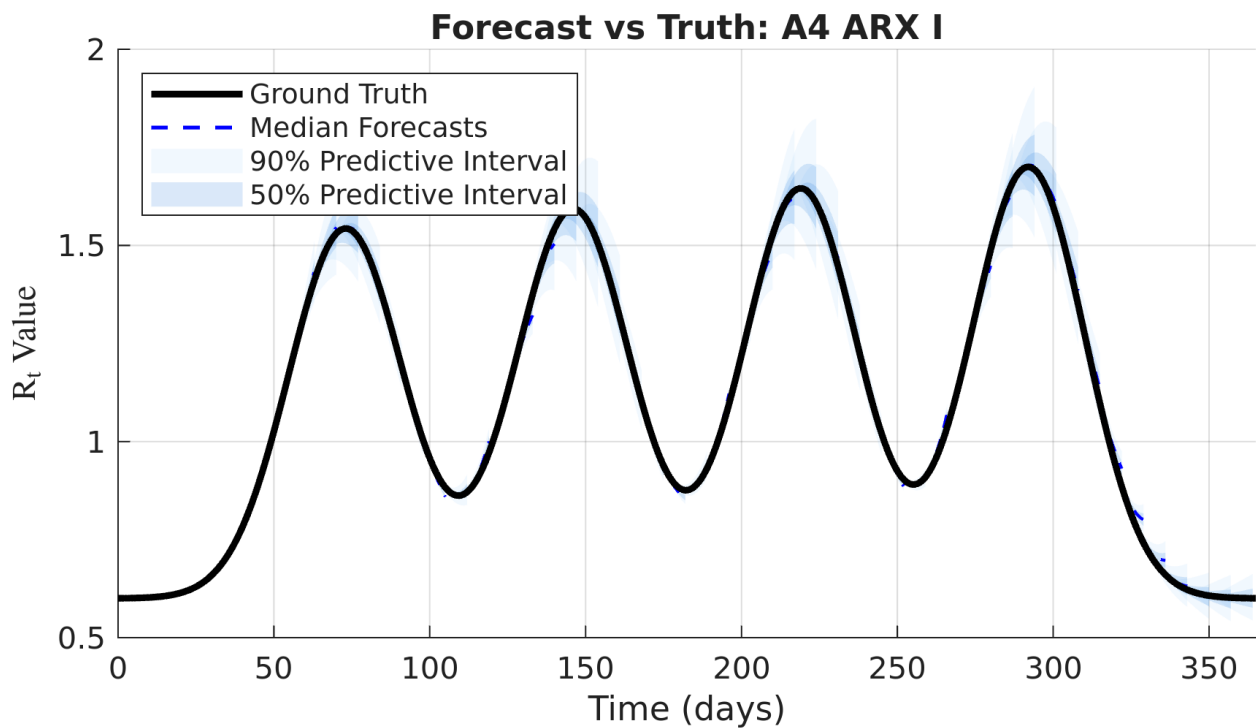


Figure 13: ARX/I on scenario A4 (amplifying resurgence). Mean window WIS = 5.89×10^{-3} .

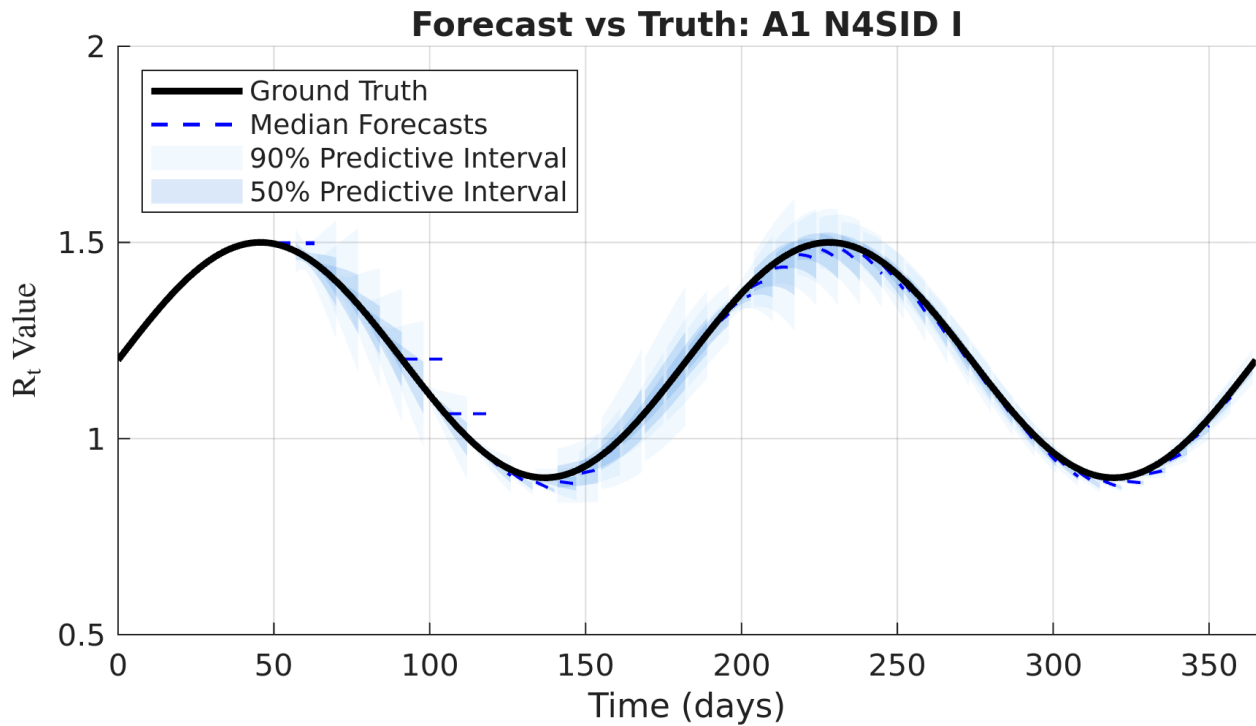
N4SID

Figure 14: N4SID/I on scenario A1 (seasonal forcing). Mean window WIS = 1.13×10^{-2} .

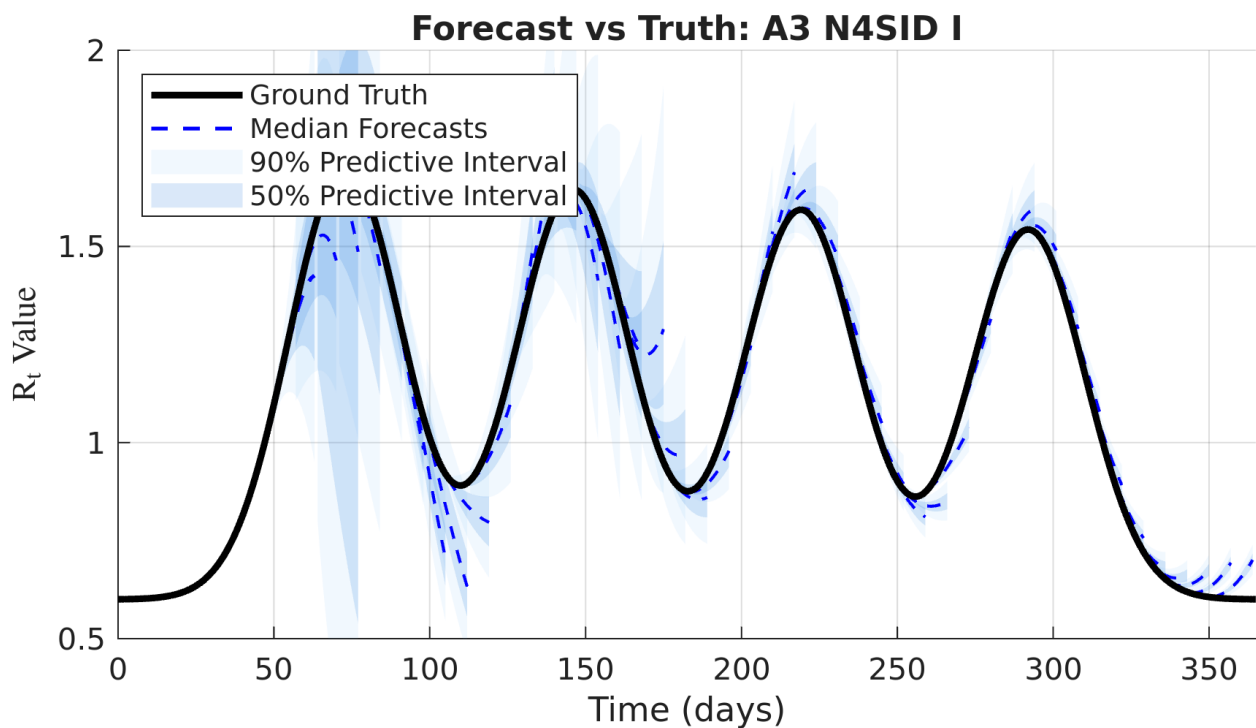


Figure 15: N4SID/I on scenario A3 (damping resurgence). Mean window WIS = 3.19×10^{-2} .

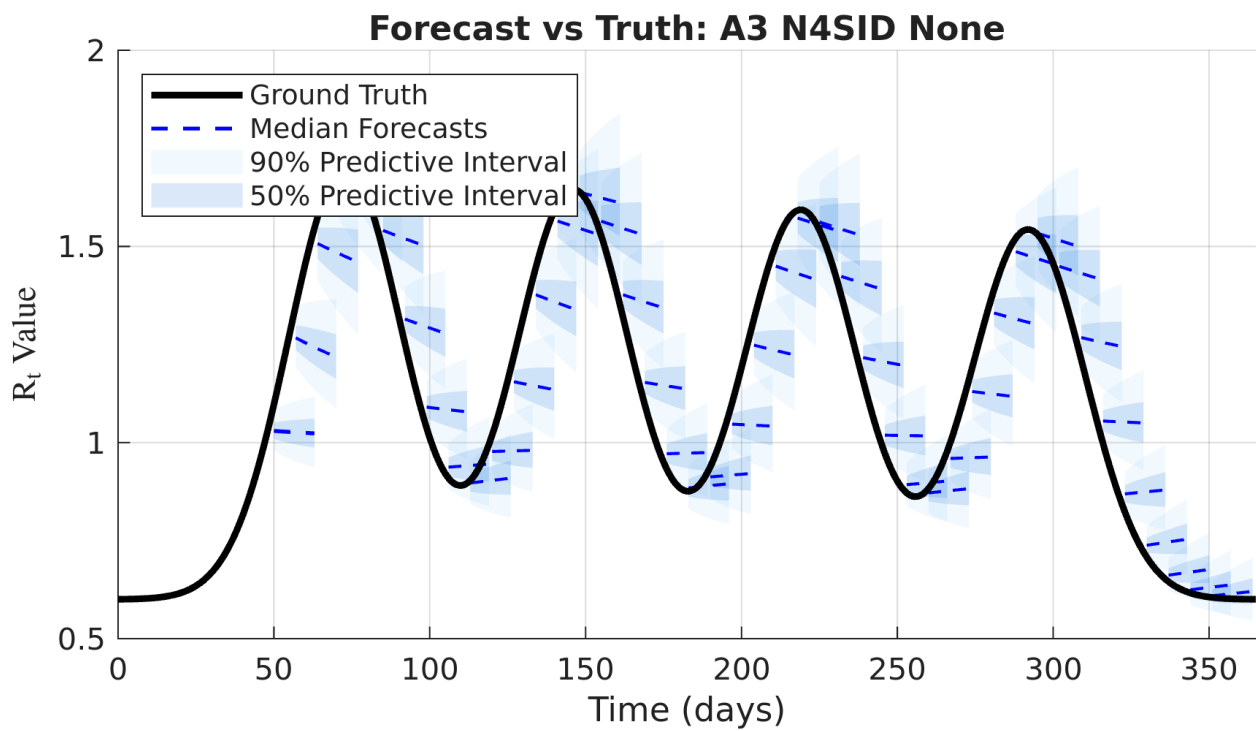


Figure 16: N4SID/None on scenario A3 (damping resurgence). Mean window WIS = 1.02×10^{-1} .

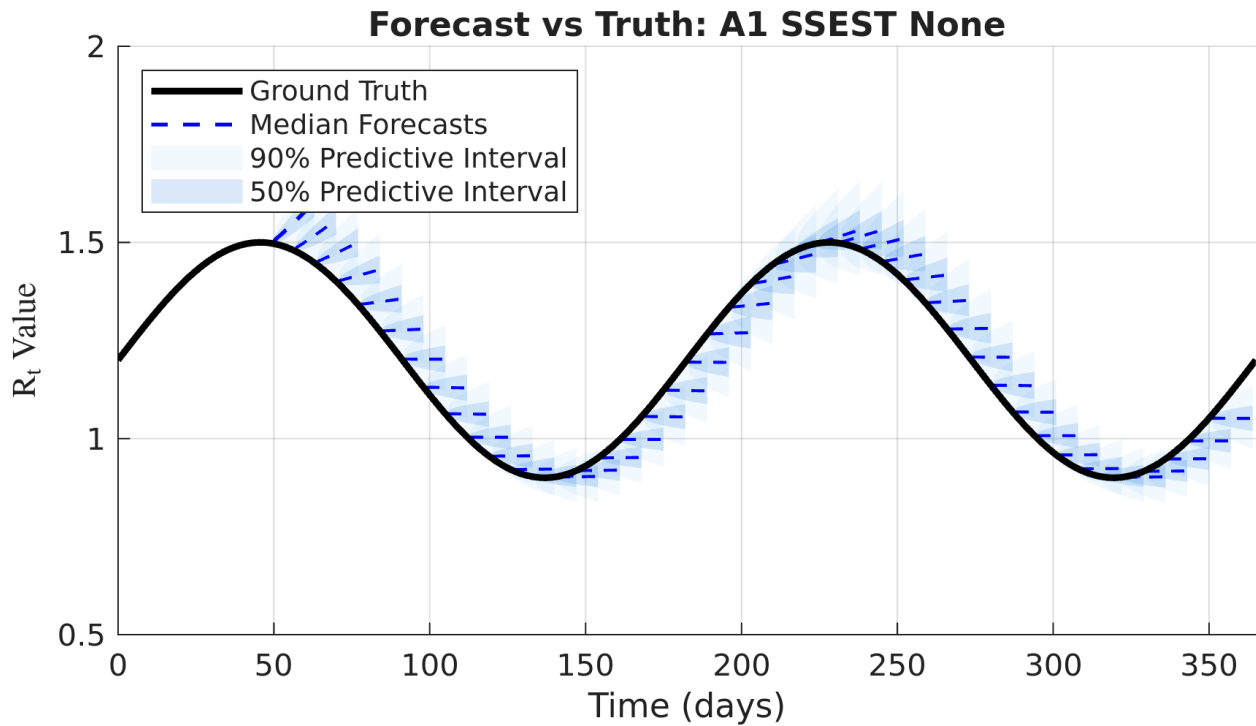
SSEST

Figure 17: SSEST/None on scenario A1 (seasonal forcing). Mean window WIS = 2.63×10^{-2} .

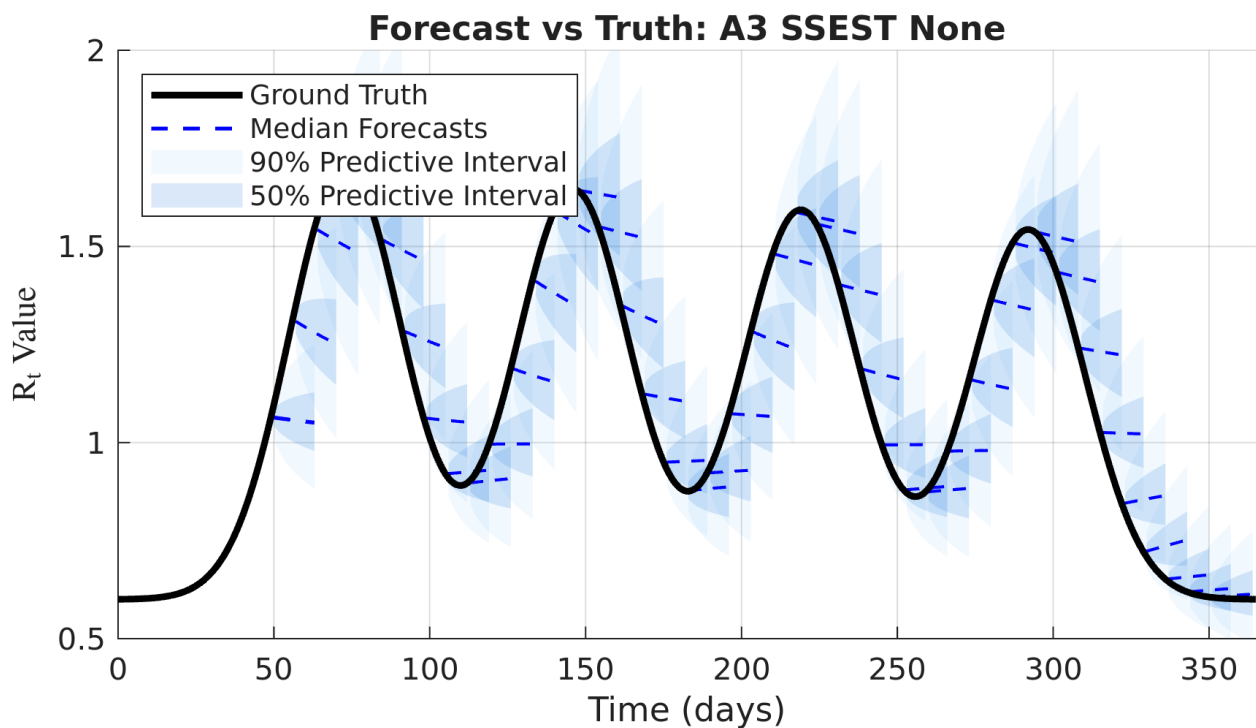


Figure 18: SSEST/None on scenario A3 (damping resurgence). Mean window WIS = 6.79×10^{-2} .

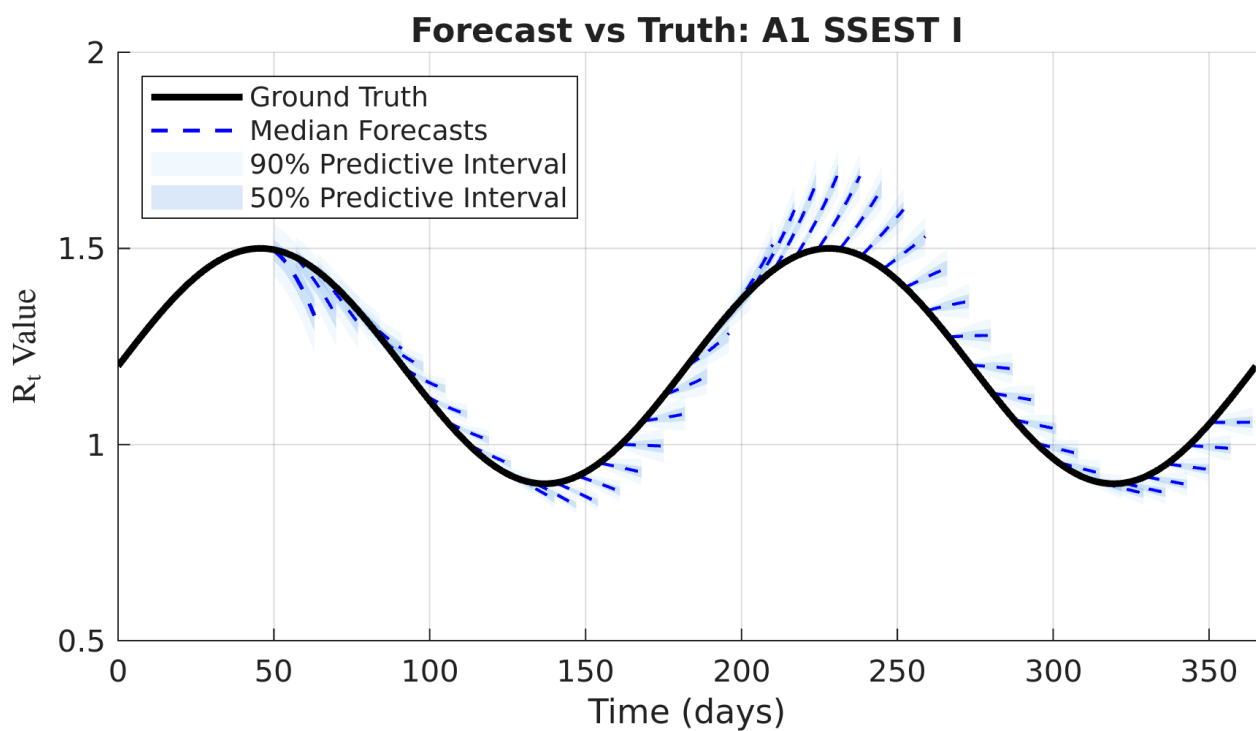


Figure 19: SSEST/I on scenario A1 (seasonal forcing). Mean window WIS = 3.33×10^{-2} .